When it Comes to Machine Preferences, Thurstone is the Model of Choice

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Abstract

Softmax output layers suggest, almost by construction, that large language models should obey Luce's Choice Axiom. They do not. In masked-token experiments a Thurstonian contest model predicted BERT's revealed preferences better in roughly 80% of cases. We extend the test to modern models across eleven designs, five models and 6,654 cells scored by exact log probabilities. Replaying the original two-slot stimuli over 2,183 cells, Thurstone beats Luce decisively (mean ΔKL +0.55, CI $[+0.51, +0.60]$), as it does over 1,195 deletion cells and under all four ordered-selection framings. The advantage rises monotonically with model tier. Explicit menus behave differently: near-deterministic, no red-bus substitution, and regularity violations beyond any random utility model.

1 Introduction

Large language models can be interrogated cheaply and at scale, which makes the study of their revealed preferences attractive whether one regards them as incipient minds or as statistical mimics of our own predilections. Either way, a question from the psychology of choice arises immediately: when a model distributes probability over alternatives, what law governs how that distribution changes as the set of alternatives changes?

The workhorse answer in economics and psychology is Luce's Choice Axiom [5], which asserts that the probability of selecting item i from a set S is proportional to a fixed utility $u(i)$:

$$P(i \mid S) = \frac{u(i)}{\sum_{j \in S} u(j)}. \tag{1}$$

Its signature is Independence of Irrelevant Alternatives: the ratio $P(A)/P(B)$ is unchanged by the presence or absence of a third option C.

There is a natural alignment between this assumption and the token probabilities of language models, which invariably emerge from a softmax:

$$p_i = \frac{e^{z_i}}{\sum_{j=1}^{n} e^{z_j}}, \tag{2}$$

however complicated the transformer computation producing z_i may be. Delete item 1 from consideration by setting $z_1 = -\infty$, a surgical procedure impossible in humans, and the remaining probabilities are the old ones renormalized by $1/(1-p_1)$. In using softmax layers we are, in effect, heavily *suggesting* to the machine that it obey Luce's Choice Axiom.

It doesn't. This paper demonstrates that language models, despite the suggestion built into their output layer, exhibit preference behavior better modeled in the Thurstonian framework of noisy contests [7].

The evidence is not a single experiment. Table 1 lists eleven designs comprising 6,654 scored choice cells, built from 10,704 logged exact log-probability measurements on five GPT models, alongside the masked language model study of the 2024 version and a generative replication on Claude. Every cell is a zero-parameter out-of-sample prediction: both families are calibrated to an unrestricted distribution and asked to predict a restricted one they have not seen. The two largest batteries, 3,660 and 1,195 cells, favor the contest model with bootstrap confidence intervals excluding zero on every one of the five models, and the margin rises monotonically with model tier, from +0.30 on GPT-4.1-nano to +0.88 on GPT-4.1. All raw model responses are logged and released, so every number below is recomputable without further inference.

Section 2 states the two competing models. Sections 3–4 test them on masked language models. Section 5 moves to modern models under exact measurement, using polysemous words as the entry probe and connecting the paradox to the stated-versus-generated gap recently documented by Cekinmez, Wu, Marjieh and Griffiths [8]. Section 6 draws out the implications: a mechanism for generative "mode collapse," a principled diversity knob, and a case for Thurstonian layers.

Design	Models	Cells	Verdict
Adjective qualification (masked)	2 masked LMs	100+	Thurstone $\approx$ 80%
Two-slot elimination (masked)	2 masked LMs	100+	Thurstone $\approx$ 80%
Polysemous generation, restriction	Claude	21	re-collapse, not renormalization
Qualified-noun restriction	3 GPT	71	tie (degenerate priors)
Random elicitation, one deletion	3 GPT	77	Thurstone 39/56, $p = 0.002$
Permutation-controlled deletion	5 GPT	1,195	ΔKL +0.025 [+0.018, +0.033]
Original 2024 stimuli, two-slot	5 GPT	3,660	ΔKL +0.52 [+0.49, +0.55]
Ordered selection to depth five	5 GPT	1,363	+0.23 to +0.41, all four framings
Reversibility (best vs worst first)	5 GPT	108	both families fail; not one scale
Block–Marschak / RUM test	5 GPT	30	30/30 violate RUM
Menus, duplicates, decoys	5 GPT	110	no substitution; 16/30 regularity

Table 1: The full evidence base. Cells are scored zero-parameter predictions of a restricted distribution; GPT cells use exact log probabilities rather than sampling. Masked language model counts are category/adjective combinations from the 2024 version.

2 A choice of choice models

One might describe Equation (1) as an *urn model*: item k is represented by balls in an urn in proportion to $u(k)$, a choice is a random draw, and shrinking the choice set merely removes balls. The renormalization of probability under restriction then seems perfectly reasonable.

But perhaps preference is a horse race, not an urn. In the competing model each item i carries a location parameter a_i (a dislikability, say) and an associated performance $X_i \sim N(a_i, 1)$. Item i is chosen when its performance is the lowest draw. This is the model of Thurstone [7], and it makes different predictions about how choice probabilities respond when the choice set is altered: rather than removing a ball from the urn, we remove a horse from the race.

(As an aside, it *is* possible to engineer a contest that satisfies Luce's Choice Axiom, by taking $X_i \sim \mathrm{Exp}(h_i)$ with common location and differing hazard rates, or, by Yellott's theorem, with Gumbel noise [16]. The comparison here can therefore be viewed as between two noise families within the contest framework, which is why single-menu fits cannot separate them and the restriction, duplication, and context designs below are required.)

The practical obstacle to Thurstonian models has always been computation, not plausibility. A fast lattice algorithm for calibrating multi-entrant Thurstone models, recovering locations $\{a_i\}$ from winning probabilities and computing derived quantities for fields exceeding 100,000 entrants, was given in [1], and is what makes the experiments below routine.

3 Experimental design: masked language models

We probe the token probabilities of BERT [2] and RoBERTa [4], prompted to fill in blanks that invite choices among items of a category. Two designs are used.

In the first, the model answers an *original* question and a *qualified* question whose only difference is an adjective that shrinks the choice set:

1. *Original:* "My favorite state in the U.S. is [MASK] and I try to visit once a year."
2. *Qualified:* "My favorite Western state in the U.S. is [MASK] and I try to visit once a year."

In the second, we eliminate a single option explicitly, feeding the top answer of a first prompt back into a second:

1. *First:* "My favorite primary color is [MASK] because it is SOMETHING."
2. *Second:* "My two favorite primary colors are [ANSWER] and [MASK] because they are SOMETHING."

where SOMETHING ranges over many adjectives so that luck of phrasing is averaged out (the appendix of the working version lists roughly one hundred category/adjective combinations).

Both models make zero-parameter predictions of the restricted distribution from the unrestricted one. Luce's prediction is renormalization, Equation (1); in the two-step design this is the simplest Plackett–Luce [6] or Harville [3] construction. The Thurstonian prediction calibrates locations $\{a_i\}$ to the unrestricted token probabilities via $\mathrm{Prob}(X_i < X_k \ \forall k \neq i) = p_i$, removes the excluded contestant, and recomputes winning probabilities.

4 Results for masked language models

Table 2 shows the "favorite Western state" prompt for BERT. The Thurstonian column tracks the actual restricted probabilities more closely than Luce renormalization, which overweights the favorite (California) at the expense of the field. Table 3 repeats the exercise for "favorite girl's baby name," where the RMSE using Luce's probabilities is roughly 50% higher.

Across the full panel of qualified-question and single-item-elimination experiments, the Thurstonian model provides the more accurate prediction in approximately 80% of cases, with its advantage concentrated where the qualification is substantial and the entropy of the unrestricted distribution is low.

State	Original (%)	Luce (%)	Thurstonian (%)	Actual (%)
California	10.89	29.87	25.38	20.17
Arizona	6.74	18.50	17.20	10.00
Texas	5.50	15.10	14.59	7.68
Colorado	3.18	8.74	9.38	7.81
Oregon	2.89	7.92	8.67	8.57
Oklahoma	1.97	5.41	6.37	6.64
Nevada	1.66	4.55	5.54	10.18
Montana	1.59	4.37	5.36	14.27
Idaho	1.05	2.87	3.82	4.99
Wyoming	0.98	2.69	3.62	9.69

Table 2: Predicted probabilities under Luce's Choice Axiom and the Thurstonian model for the "favorite Western state" prompt (BertForMaskedLM). Both predictions use only the unqualified token probabilities; the Thurstonian model is closer to the actual qualified probabilities.

Name	Original (%)	Luce (%)	Thurstonian (%)	Actual (%)
Lily	1.56	20.35	16.18	14.52
Emily	1.15	15.06	13.31	16.29
Mia	0.86	11.25	11.02	13.18
Bella	0.85	11.12	10.93	8.84
Sarah	0.68	8.89	9.44	9.41
Rachel	0.59	7.66	8.56	7.15
Kate	0.59	7.63	8.54	8.44
Lauren	0.47	6.11	7.39	7.23
Brittany	0.46	6.04	7.33	7.55
Chloe	0.45	5.89	7.21	7.39

Table 3: The same comparison for "favorite girl's baby name." The RMSE using Luce's probabilities is approximately 50% higher than the Thurstonian RMSE.

5 Modern models under exact measurement

Cekinmez, Wu, Marjieh and Griffiths [8] recently turned an everyday property of language into a measurable distribution over meanings: give a model a bare polysemous word ("bolt") with no disambiguating context, sample many outputs, and classify which sense each expresses. Across 32 text and image generation models they report a striking ladder of normalized sense entropies: image generation 0.10, text generation 0.25, human subjects 0.47, and, when models are asked to *predict* the distribution of senses, roughly 0.73. Models know the ambiguity; they do not express it. The finding sits within a growing literature on generative homogenization [9] and its sharpening under preference tuning [10]; their own ablation shows DPO alone cutting sense entropy from 0.25 to 0.18.

Read through the lens of Section 2, the ladder is what a contest model produces. Let the stated distribution proxy the latent locations $\{a_i\}$; let each generation be a contest among noisy sense performances. When noise is small relative to the spread of locations, winning probability is a violently convex function of location differences, and a mildly uneven prior collapses onto its mode. One noise parameter then indexes the entire ladder, and the "multimodal gap" becomes a noise gap rather than a knowledge gap. The rival explanation is a power tilt: renormalized p_i^γ, the Luce family with a temperature, which is mechanically what RLHF sharpening, low-temperature decoding, and classifier-free guidance perform. The two families are distinguished not by the collapse itself but by behavior under choice-set restriction, precisely the instrument of Section 3, and an experiment run by neither paper until now.

5.1 Replication on a chat model

The smallest of our designs, and the only one using sampling rather than exact probabilities, replicates their generative setup on Claude models (October 2025 generation): 15 polysemous words from the stimulus set of [8], 150 independent sentence generations per word, senses classified by a second model against the closed inventory, plus a stated distribution elicited with their "predicting itself" framing. The collapse replicates almost exactly: mean normalized sense entropy 0.252 generated (their text-model figure: 0.25) against 0.880 stated.

Fitting the two one-parameter families from stated to generated distributions gives a statistical tie (Thurstone $\sigma = 0.42$, RMSE 0.273; power tilt $\gamma = 2.25$, RMSE 0.273; Thurstone the better fit on 10 of 15 words), and the reason both fit poorly is itself informative: for several words the generated mode is not even the stated argmax ("port" generates *harbor* in every sample while rating *connector* most probable). Stated reports are a biased window on the latent locations, not merely a flattened one. This is consistent with [8], whose stated distributions overestimate even human diversity. Calibration should therefore use revealed behavior, which is what the restriction design does.

5.2 Choice-set restriction

For the six words whose unrestricted distributions were non-degenerate, we re-sampled with the modal sense excluded by instruction (150 samples per condition) and compared zero-parameter predictions of the restricted distribution: Luce renormalization against Thurstonian contestant removal, both calibrated to the add-half smoothed unrestricted sense counts.

Word	Luce RMSE	Thurstonian RMSE	Restricted distribution
bolt	0.450	**0.413**	fastener .68, run .18, lock .14
seal	**0.077**	0.089	animal .67, close .33
pitch	0.655	**0.630**	throw .06, tone .31, field .63
bat	0.559	**0.546**	animal 1.00, hit .00
match	0.758	**0.651**	firestick .82, contest .18
iron	**0.002**	0.037	metal .95, press .04, golf .01
pooled	0.495	**0.459**	

Table 4: Restriction test on polysemous words (Claude Haiku 4.5, 150 samples per condition, modal sense excluded by instruction). The Thurstonian prediction is closer pooled and on four of six words (bootstrap over words: $P(\text{Thurstone better}) = 0.90$).

The margin favors Thurstone, but the qualitative signature matters more than the margin: under restriction the model does not spread probability proportionally, as Luce requires. It *re-collapses* onto a new modal sense (match → *firestick* at .82; iron → *metal* at .95; bat → *animal* at 1.00). Remove the winning horse and the runner-up now wins nearly always. That is low-noise contest behavior, in the same low-entropy, heavy-qualification regime where the Thurstonian advantage concentrated in Section 4. Some rank flips under restriction exceed even the Thurstonian sharpening, and inventories with overlapping senses ("bat") contaminate the exclusion instruction; neither family survives those cells.

5.3 Exact probabilities on GPT models

Sampling noise can be removed entirely on models that expose token log probabilities. We ported the qualified-question design of Section 3 to GPT-4o-mini, GPT-4o and GPT-4.1: twenty category pairs ("my favorite color is" / "my favorite warm color is"), two phrasings, next-token distributions read exactly from the top-20 log probabilities, with the Thurstonian field restricted to a declared category inventory. Across 71 usable cells the comparison is a statistical tie with a Thurstonian lean: Thurstone wins 38 of 71 cells and has the lower pooled RMSE on all three models (e.g. 0.290 vs 0.295 on GPT-4o), but a bootstrap over cells does not separate the families ($p = 0.29$).

The instrument, however, has weakened for an interesting reason: the unqualified distributions of preference-tuned chat models are already near-degenerate ("my favorite tree is" puts 0.94 on *oak*), so most cells carry little discriminating information. Under either model the surviving favorite keeps nearly everything, and the comparison hinges on tail probabilities. The masked models of Section 4 answered the unqualified question with entropy to spare (California at 10.9%); their preference-tuned descendants have already collapsed before the experimenter arrives. This is the same collapse measured by [8], now visible as a loss of experimental power: alignment has pushed the interesting variation into the tails, where only exact log-probability designs can reach it.

5.4 Random elicitation restores the regime

The remedy is to ask for randomness rather than preference. "Name a random programming language" followed by explicit single-item elimination raises mean unqualified entropy to 0.38 and returns the experiment to the regime where the two families disagree. The elimination takes one of two forms, either an exclusion ("… that is not Python") or the two-slot continuation of Section 3 ("I already picked one random programming language: Python. Name another…"). The recovery is only partial, since models are poor samplers even on request, with "a random planet" putting essentially all mass on *Mars*. The homogenization of [9] extends to explicit randomness.

To control for phrasing and for which item is removed, the definitive battery deletes *each* of the top eight observed items in turn, under two phrasings, across 50 categories and three GPT models. Thirty-one categories retain enough open-vocabulary mass to be scorable, giving 692 permutation-controlled cells, scored by the proper rule KL(actual $\|$ prediction), with bootstrap confidence intervals over cells.

Contest behavior governs wherever machine preference is genuinely stochastic; the Choice Axiom is competitive only in the degenerate cells that preference tuning has emptied of choice.

Stratum	n	mean ΔKL (Luce − Thurstone)	95% CI
all cells	692	$+0.025$	$[+0.016, +0.035]$
non-degenerate prior ($\tilde{H} > 0.2$)	521	$+0.036$	$[+0.026, +0.048]$
degenerate prior ($\tilde{H} \leq 0.2$)	171	-0.008	$[-0.025, +0.011]$

Table 5: Permutation-controlled deletion battery, exact log probabilities, KL scoring. Positive favors Thurstone. The Thurstonian advantage is decisive overall and concentrates precisely where the model retains genuine choice entropy.

5.5 The original stimuli at scale

The largest battery, and the largest effect, comes from re-running the *original* stimulus set of this paper's 2024 version. That is 97 scorable categories crossed with every adjective in the original appendix, 852 distinct prompt pairs, each put to five GPT models in the original two-slot form ("My two favourite human organs are the [ANSWER] and the [MASK] because they are SOMETHING") with exact log probabilities. Across the 2,183 cells of the three headline models the conditional second choice decisively favors the contest model: mean ΔKL $+0.55$, 95% CI $[+0.51, +0.60]$, and in both entropy strata. Adding the two cheaper models brings the battery to 3,660 cells and $+0.52$ $[+0.49, +0.55]$. Predicting the runner-up given the winner is precisely where Luce renormalization fails hardest. It is the machine-preference analog of the Harville [3] overpricing of exactas that motivated the Thurstonian computation in the first place. The same questions asked of BERT in 2024 and of GPT-4-class models in 2026 return the same verdict, with a larger margin under exact measurement.

Across the five models the margin is ordered by tier: GPT-4.1-nano $+0.30$, GPT-4o-mini $+0.35$, GPT-4o $+0.42$, GPT-4.1-mini $+0.65$, GPT-4.1 $+0.88$. The more capable the model, the further its revealed preferences sit from the Choice Axiom its own output layer suggests. Two readings are available and this design does not separate them. Capable models may represent preference in a way better described by a latent contest, or they may simply hold sharper locations, since the deficit also grows with the concentration of the unrestricted distribution. The second reading is the more conservative, and the degenerate stratum of Table 5 is consistent with it. What the ordering does establish is that the effect is not an artifact of weak models, which is the usual objection to results of this kind. Whether Thurstonian diagnostics can serve as a label-free measure of model quality is a natural question, but Section 5.6 counsels caution: the reversibility diagnostic does not order the models by capability at all.

One further caution emerged with practical import for all such experiments: the "actual" restricted distribution depends materially on how the restriction is phrased. Holding everything else fixed, the exclusion and two-slot variants disagree about the restricted distribution by a median maximum gap of 0.14, and up to 0.65 (GPT-4o-mini names *Earth* after *Mars* with probability 0.94 under one phrasing and 0.56 under the other). Under surgical removal from a fixed preference state the two phrasings would coincide; in a language model the choice-set manipulation is itself a prompt intervention. Framing effects of this size sit above both choice models, and any account of machine preference that ignores them is incomplete.

5.6 Ordered selection, depth, and reversibility

The two-slot design is an exacta. Asking instead for an ordered list, and eliciting each slot separately with the earlier picks fed back in, extends it to a trifecta and beyond, so that Plackett–Luce [6] renormalization and Thurstonian contestant removal can be compared at every depth. Four framings were run over 40 categories and five models to depth five: two neutral ("select five random X in order," "name five different X, in order") and two preference-ordered (most favourite first, least favourite first).

The contest model wins under all four framings, which is the robustness that matters most here, since the framings are different elicitation architectures rather than paraphrases: +0.55 most-favourite-first, +0.38 "different," +0.26 "random," +0.18 least-favourite-first (all intervals excluding zero, 1,363 cells).

Depth, however, does not behave as the racing analogy predicts. Since slot d involves $d - 1$ removals and renormalization can only redistribute vacated mass proportionally, the Luce deficit might be expected to compound with depth. It does the opposite, falling from +0.42 at slot two to +0.24 at slot five, and within a given category and model the change from the second slot to the deepest is negative on average (−0.23, positive in only 161 of 395 ladders). The mechanism is visible in the cell sizes: each removal shortens the surviving field, and by slot five most cells have only two or three candidates left, where the two families cannot differ much. Depth in an open-vocabulary elicitation removes discriminating power rather than adding it, which is the opposite of the racing case, where the field stays large as places are filled.

Reversibility is a sharper test, and both families fail it. Under a Thurstonian scale, asking for the least favourite is the same latent vector read backwards, since the minimum of X is the maximum of $-X$; one calibration must then predict both directions with no free parameters. Luce must instead invert its utilities. Neither works: predicting the least-favourite-first distribution from the most-favourite-first one costs 5.3 to 9.0 nats depending on the model, for both families alike, and the ranking of models by this quantity does not track capability. The reason is that the two directions barely share an item set. Asked for a least favourite, models name items they never mention when asked for a favourite, so the reversed question is not the same contest re-read but a different one. Machine preference in this regime is not a single scale with two ends.

5.7 Menus, duplicates, and context effects

Presenting the choice set explicitly changes the phenomenon entirely, in three ways. The design is "pick one of these: ...," with menu order randomized and probabilities read from option-token log probabilities.

First, menu choice by preference-tuned models is nearly deterministic. A transport menu (car, bus, train, bike, walk, taxi) under contextual conditioning behaves with perfect semantic sense. "It is raining" sends GPT-4o to *taxi* with probability 1.00 and "there is a fuel shortage" sends it to *bike* at 0.99. But the resulting distributions are so degenerate that menu-based deletion cannot separate the choice models (144 cells, ΔKL CI $[-0.028, +0.016]$), consistent with Table 5's stratification. The determinism extends to simulated agents: fictional election candidates with platform descriptions, evaluated through fictional voter personas, produce near-certain votes even when the persona is written to be genuinely cross-pressured, and candidate-withdrawal cells are consequently uninformative. The model represents a persona as an argmax decider; it does not carry within-persona randomness at all.

Second, machines are anti-red-bus. In Debreu's classic objection to the Choice Axiom [11], splitting a bus into a red bus and a blue bus should not double the bus vote; humans treat near-duplicates as substitutes, which correlated-noise (probit) models capture. Across ten duplicate-pair families the models show *no substitution at all*: the duplicated category's total share typically inflates, often beyond even the urn-doubling prediction (apple 0.08 → 0.53 when split into green and red), and a substitution baseline is the best predictor in only 15 of 41 cells against Luce's 24. (With exchangeable duplicates, Luce and independent Thurstone nearly coincide; the test's power is entirely against substitution.) Description salience, in that a *red* apple is more vivid than an apple, appears to dominate any shared-noise structure.

Third, where a menu does leave residual entropy, machine choice violates *regularity*, the property, shared by every random utility model including Thurstone's, that an added option cannot increase an incumbent's absolute share. Using two-attribute options (price versus quality) in randomized lettered menus, an asymmetrically dominated decoy [12] increased the target's absolute share in 16 of 30 family-model cells, and a compromise manipulation [13] in 7 of 30. The phone-plan family shows a textbook attraction effect (GPT-4o: target share 0.26 → 0.84 when its decoy appears). Machine preference, like human preference, is not a random utility model in the menu regime.

The menu regime admits a verdict stronger than any pairwise model comparison, because exact probabilities make it feasible to measure the *complete choice structure*: all eleven menus over a four-item universe, a measurement no human experiment delivers cleanly. By the Block–Marschak–Falmagne theorem [14, 15], such a structure is random-utility representable if and only if its Block–Marschak sums are nonnegative. Across six item families and five GPT models, **all 30 measured structures violate the conditions** (4–9 negative terms of 32 each; worst −0.77, far beyond measurement noise from order averaging). Fitting the structures anyway, the best possible random utility model (a distribution over all 24 rankings) leaves half the divergence unexplained (total KL 8.1, against 16.2 for the best Thurstone locations and 17.5 for the best Luce utilities, with Thurstone the better parametric fit in 26 of 30 structures). In the menu regime, machine choice is not a random utility model at all.

Together the batteries locate the choice law in the *elicitation architecture*. Open-vocabulary recall behaves like a Thurstonian contest among retrieved candidates (Table 5); explicit menus behave like near-deterministic maximization perturbed by salience and context effects that no random utility model accommodates. Between them sits the sampled-sentence regime of [8], where the contest reading again does best.

6 Discussion

To the extent that LLMs mirror our own preferences, these results provide a rejection of Luce's Choice Axiom in favor of the Thurstonian model. The popularity of the former may owe much to computational convenience, a consideration the algorithm of [1] removes.

The polysemy results suggest something further: that generative "mode collapse" is not a defect of knowledge but the lawful output of a low-noise contest run over intact locations. That reframing carries a practical remedy. If choice is a contest, diversity is restored not by a power tilt but by inverting the contest to recover locations and re-sampling at a chosen noise level. Raising the sampling temperature distorts the tails and does not renormalize the way behavior actually renormalizes. The calibration is exactly the fast Thurstone computation of [1], and it offers a principled diversity knob without retraining.

Leaning into the irony of softmax machines disobeying Luce's Choice Axiom, the results also suggest that Thurstonian neural network layers might generalize better than softmax layers, the engineering challenge notwithstanding. Current architectures force hidden layers to fight the renormalization behavior their output layer implies; a contest-structured output layer would not need to.

Finally, this study is coarse-grained, and the pilot of Section 5 is small: fifteen words, one generation model, a single machine judge, sampling temperature not controlled. The full stimulus set and model panel of [8] would settle the one-parameter comparison with real power, and any model of machine decision-making that satisfies Luce's Choice Axiom might, on this evidence, be viewed with suspicion, and readily improved.

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